Group Need-to-Know Guide

Smart lighting and plant monitoring • concise oral-exam and demonstration briefing

The project in 30 seconds

We use two Arduino Uno WiFi Rev2 boards. One reads sofa activity, screen position, light and touch, then uploads to ThingSpeak every 20 seconds. A Mac reads the cloud data and controls Hue lights. The other Arduino uploads raw soil moisture independently. We demonstrated the lighting node through a phone hotspot while the Mac and Hue lights stayed at home. There is no automatic watering.

Know these connections and numbers

Item	Remember
Lighting inputs	A0 sofa; A1 screen; A2 light; D2 touch
Plant input	A0 on a separate board; SIG / 5V / GND
Thresholds	Sofa >100; screen >200; bright ≥500; touch HIGH
Intervals	Lights upload 20 s; Mac polls 5 s; plant uploads ~60 s
Data age limits	Lighting 75 s; plant display 180 s
Timers	Activity hold 30 min; manual override 15 min
Automatic hours	07:00–23:00, Europe/Oslo
Brightness	WHITE 100%; WARM 50%; CINEMA 25%, approximate

Table 1. Final demonstrated configuration.

Understand the priority order

First require fresh light data. If data are missing or stale, keep the actual lamp state. With fresh data: screen down wins and cancels manual override; otherwise manual override wins; otherwise recent sofa/touch activity gives warm light; otherwise dark during 07–23 gives white light, and all other conditions give off.

Removing an object from the sofa does not switch the light off immediately: activity remains for 30 minutes. Removing an object from the screen sensor ends cinema mode and returns to the next applicable rule. Sofa above its threshold continuously renews activity on each new measurement.

Which code does what?

File / component	Responsibility
Lighting Arduino sketch	Reads sensors, latches touch, reconnects Wi-Fi, uploads four fields
Plant Arduino sketch	Reads moisture A0 and uploads one field; no lighting decisions
thingspeak_lights.py	HTTPS read; field/range validation; cloud age and threshold conversion
thingspeak_controller.py	Polls; accepts each fresh entry once; updates Hub; starts controller, optional TalkBack and web
lighting.py	Presets, 30 min activity hold, freshness gate and automatic policy
hub.py	Shared lock, policy state, 15 min manual override, dashboard status
control_hue.py / run_controller	Send changed desired states to real Hue lamps; record errors
talkback.py	Existing cloud command consumer; final stale-command handling needs verification
thingspeak_moisture.py	Independent raw moisture and timestamp/freshness; no percentage
webapp.py + dashboard.html	Authenticated local UI; reads moisture server-side; displays status

Table 2. Explain responsibilities rather than memorising every line.

Important code ideas

```
motion = int(sofa > 100 or touch == 1)
screen_down = int(screen > 200)
light_bright = int(light >= 500)
```

Raw analog readings are integers in the accepted 0–1023 range. Thresholds turn them into logical states. They are not precise distance, lux or moisture-percentage measurements.

Each ThingSpeak entry has an ID and UTC creation time. Processing the same entry repeatedly would incorrectly renew the activity timer. The controller skips duplicate entries and subtracts measurement age from the monotonic clock before updating policy. Monotonic time measures elapsed durations without depending on wall-clock adjustments.

A shared lock prevents concurrent web, TalkBack and sensor operations from modifying Hub state at the same time. The light-output worker sends a Hue request only when the desired preset changes; failures are exposed in dashboard status.

Common questions and short answers

Why two Arduinos?

The plant node can stay by the plant while the lighting node travels to school. Their channels and roles are independent.

Why keep a Mac running?

It bridges cloud measurements/commands to the home Hue Bridge. The Arduino does not directly drive lamps.

Does the Arduino work without its programming PC?

Yes, with USB power and Wi-Fi. The Mac controller is still required for this lighting architecture.

Why can a response take several seconds?

Uploads are every 20 s and reads every 5 s. Roughly 0–25 s of scheduling delay is expected, plus network time; it is not a guaranteed bound.

What happens if the internet stops?

No fresh light measurements arrive. After 75 s the policy issues no new output and leaves the physical lamp state unchanged. A complete outage/recovery test still needs evidence.

Can we open localhost from school?

No. 127.0.0.1 refers to the device running the browser. The final dashboard is local to the Mac. TalkBack provides browser-based remote commands.

Is the moisture number 598 a percentage?

No. It is a raw sensor value. No reliable dry threshold or percentage calibration was established.

Why lower the screen threshold?

The original 400 threshold was unreliable at the demonstration position. Above 200 gave the reported physical cinema response. Final mounting still needs repeated calibration.

Why no orange cinema light?

The selected Hue lights support colour temperature. Cinema uses warm white, CT 454, at approximately 25% brightness.

What is actually tested?

Cloud readings, real Hue off/activity/screen responses, hotspot operation and plant dashboard were observed. Exact latency, long timer expiry and failure cases are not yet fully measured.

Is MQTT still used?

The older repository firmware uses MQTT, but the final demonstrated path uses ThingSpeak HTTPS. Do not describe the old firmware as the uploaded sketch.

Where are secrets stored?

Locally in .env and Arduino secrets settings. Keys must not appear in browser source, Git, screenshots or submitted code.

Demo script and safe operation

• Before filming: Mac awake/on power, lid open, controller running once; phone hotspot on at 2.4 GHz; lighting Arduino powered; no secrets visible.

• Start with ordinary light-level automation during 07–23 before registering activity. Use AUTO if needed, but remember AUTO does not erase the 30 min activity timer.

• Trigger sofa → warm 50%; remove object and explain hold. Use TalkBack OFF then WHITE to show manual control.

• Trigger screen while WHITE is active → warm cinema 25%, if priority works as designed. Hold long enough for a cloud update.

• Show separate plant telemetry: raw value, UTC time, age, freshness. Explain that no pump exists.

• Allow about 25 s plus network margin per sensor action. A two-camera view or helper at home is needed if demonstrating from school.

Start / stop on the Mac

```
cd ~/Water-and-light-system
set -a
source .env
set +a
caffeinate -i .venv/bin/python thingspeak_controller.py --hue --talkback --web
```

Stop using Ctrl+C in the controller terminal. Do not run manual_demo.py or another real-Hue controller simultaneously. Open http://127.0.0.1:8765 on the Mac and use the local web password.

Troubleshooting

Symptom	First check
No response	Fresh ThingSpeak entry? Hotspot on? Arduino powered? Controller running?
Light goes off when screen clears	Expected if no activity/manual rule applies, especially outside 07–23
No web login setup	Run .venv/bin/python tools/setup_web.py locally; never share the password
Website old or unavailable	Local Mac address only; refresh 30 s; check terminal and web error
Hue fails but polling continues	Check dashboard error, home network, bridge and certificate settings
Wrong firmware in repository	Final Arduino sketches were local; collect them from the programming PC

Table 3. Practical troubleshooting. Never paste keys while asking for help.

Everyone should be able to say

“I can explain my own contribution, draw the cloud-to-Hue path, identify the four light inputs, explain screen priority and freshness, distinguish raw moisture from a percentage, and state which tests remain unverified.”